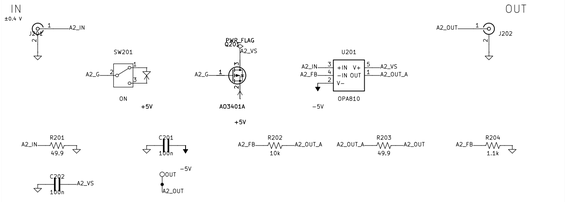
A2 x10, 14 MHz

tutorial 4.06, 5.02 (+−5 V)

cell (0,0)

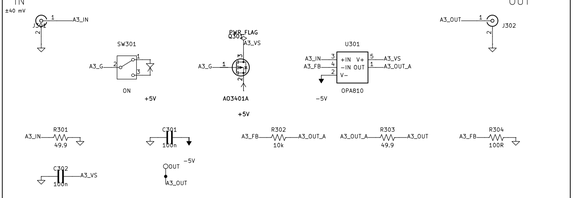


EXPECT: x10 (20 dB) into a high impedance, x5 into 50 ohm, −3 dB near 14 MHz; clips at about +−4 V
Non-inverting OPA810: gain 1 + 10 k / 1.1 k = 10.1, bandwidth 140 MHz / 10 = 14 MHz. 50 ohm at the input, 49.9 ohm
in series with the output (a 50 ohm load halves it, x5). Keep the input under 0.4 V peak: the output clips at 4 V.

A3 x100, 1.4MHz

tutorial 4.06, 5.02 (+−5 V)

cell (1,0)

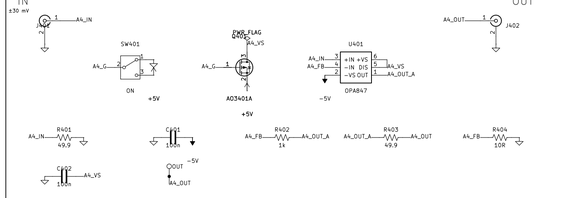


EXPECT: x100 (40 dB) into a high impedance, −3 dB near 1.4 MHz; keep the input under 40 mV peak
Non-inverting OPA810: gain 1 + 10 k / 100 = 101, bandwidth 140 MHz / 100 = 1.4 MHz. 50 ohm at the input, 49.9 ohm
in series with the output. For 40 dB out to 30 MHz use A4 (the OPA847 LNA): this tile is the FET-input, high-impedance
friendly x100 for everything slower. (The plan's two stages of x10 did not fit the cell.)

A4 LNA, 30 MHz

tutorial 4.06, 5.03 (+−5 V)

cell (2,0)

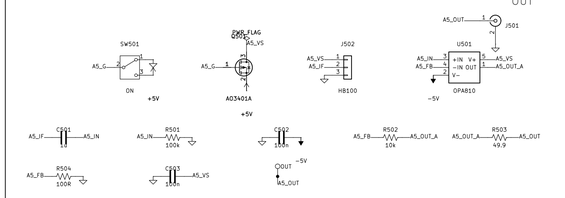


EXPECT: x100 (40 dB) from 50 ohm, flat 1–30 MHz, about 3 dB noise figure from 50 ohm; keep the input under 30 mV peak
Non-inverting OPA847 (3.9 GHz GBW at this gain, 0.85 nV/√Hz); 1 k / 10 ohm, so x101 and about 39 MHz. Its
input noise with the 50 ohm termination's own noise gives a noise figure near 3 dB. The op amp is stable only at
gains above 12 (it is decompensated): do not change Rg upward. SOT-23-6, DIS tied to +VS (enabled).

A5 Doppler, HB100

tutorial 5.08 (+−5 V)

cell (3,0)

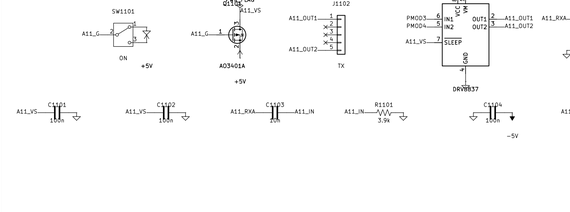


EXPECT: the Doppler beat of a target moving toward or away: 70 Hz per m/s (19.4 Hz per km/h), tens of mV after x101
HB100: a 10.525 GHz oscillator and mixer on one board; its IF is the beat between what it sends and what comes
back, 2v/lambda = 70 Hz per m/s. Wire its +5, IF and GND pins to the header: point it at something that moves.
The host's x10 input after this tile's x101 brings a hand at 1 m to the ADC's range.

A6 1-bit ADC

tutorial 7.05 (+3.3 V)

cell (0,1)

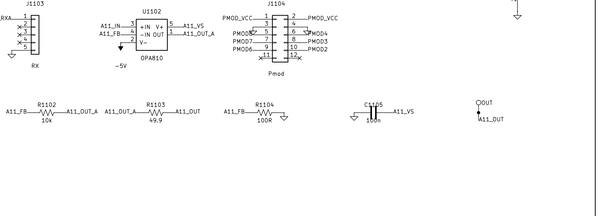


EXPECT: a one-bit stream on Pmod IO7 whose density follows IN: the FPGA closes the loop through Pmod IO8 (the handbook's gateware)
First-order delta-sigma: the comparator decides, the FPGA registers the bit and feeds it back through 10 k into the
10 k / 10 nF integrator (1.6 kHz; clock the loop at 1–10 MHz). IN takes 0 to 3.3 V, or +−1.5 V about the 1.65 V reference
if you AC-couple. Both Pmod lines are 3.3 V logic: the tile runs on the 3.3 V rail.

A7 PLL 4046

tutorial 4.01 (+5 V)

cell (1,1)

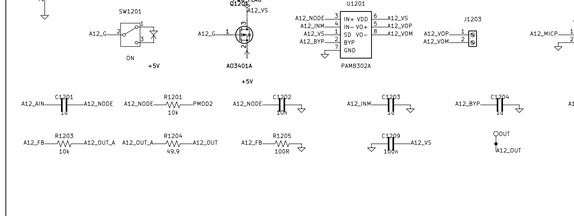


EXPECT: VCO OUT locks to a 50–200 kHz square or sine on IN; pin 9 is the demodulated FM, pin 13 the phase detector
74HC4046A: R1 10 k, C1 1 nF put the VCO near 100 kHz (R2 open: it can go down to 0). PC2 (the edge-triggered
tri-state detector) through 10 k / 100 n into VCOin: a lag filter, so expect ringing on lock. IN self-biases
through 100 n / 1 k (any level above 0.4 V peak). VCO OUT is 5 V CMOS through 470 ohm. Probe pins 9 and 13.

A8 heat

tutorial 5.02, 7.06 (+5 V)

cell (2,1)

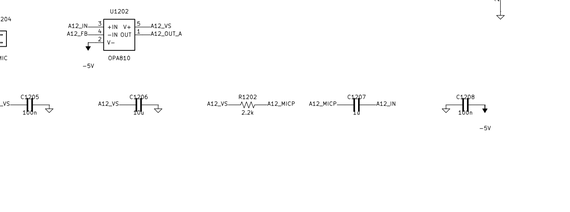


EXPECT: 0.83 W into the board when Pmod IO6 is high (the FET's drain), 10 mV/K + 500 mV on OUT
Two 15 ohm in series: 30 ohm, 170 mA, 0.83 W from the switched 5 V when Pmod IO6 is high (10 k keeps it off otherwise).
The TMP235 reads 750 mV at 25 C and climbs 10 mV per kelvin. A feedback-control experiment: hold the TMP235 at a
set point with the FPGA's PWM. (The plan's NTC divider did not fit the cell beside the two 2512s.)

A1 power, USB-C 5 V in: +5, −5, +3.3 out

tutorial 1.01

cell (3,1)



EXPECT: +5 V from the USB-C (up to 1.5 A), −5 V at 40 mA, +3.3 V at 300 mA; the green LED says 5 V is in, the red one that −5 V is up
USB-C power only: 5.1 k on each CC asks any source for 5 V. A 2 A PTC, a 5 V TVS (clamps at 9.2 V) and 20 uF.
The LM2664 charge pump makes −5 V for the op-amp tiles (40 mA; four tiles at 10 mA): it is not switched per tile, an op
amp with its +5 V removed draws nothing. The TPS7A2033 makes 3.3 V for the comparator and the lines that meet the FPGA.
The socket along the top carries +5, ground, −5, ground, +3.3: the bench's rails for a breadboard experiment.

A9 mixer, MC1496

tutorial 1.10, 6.02 (+−5 V)

cells (0,2), (1,2)

EXPECT: IF OUT = RF x LO (sum and difference frequencies, about 0.3 V peak per volt of RF at 0.3 V of LO); a balanced mixer, the carrier mostly cancels
Gilbert cell on +−5 V: signal pair at ground (51 ohm), carrier pair at 2.5 V (51 ohm to the 10 k / 10 k bias), 1 mA
per side (3.9 k), 1.5 k loads. Keep RF under 0.3 V and LO near 0.3 V peak (above that the carrier pair just switches,
which is fine for a product detector). IF OUT is single-ended from pin 6 through 100 n: a 1.5 k source, so read it
with a high-impedance input, not 50 ohm. No carrier-null trimmer: expect −40 dB of LO feedthrough.

A10 impedance, V and I

tutorial 4.02, 4.10 (+−5 V)

cells (2,2), (3,2)

EXPECT: S OUT and X OUT (each half the node's voltage into 50 ohm); Z = R X / (S − X); +−4 V and up to 40 mA for an I–V curve
DRIVE IN +2 is S. The DIP picks R between S and X (47 ohm for LEDs and diodes, 470 for most parts, 4.7 k and 47 k for
high impedances): the part goes on the OUT jack. Calibrate with the OUT open and shorted. An LED's knee against
its colour (P12) gives Planck's constant: a crystal (P8) shows its series and parallel resonances.

A11 ultrasound

tutorial 4.06 (+−5 V)

cells (0,3), (1,3)

EXPECT: a 40 kHz burst from Pmod ID4/OD3 leaves the TX transducer; its echo arrives x100 on RX OUT (hold a reflector over the board)
Drive IN1/IN2 (Pmod ID4, ID3) in antiphase at 40 kHz for 8–20 cycles: the bridge puts 10 V peak to peak across the TX
transducer. The RX transducer's 10 nF / 3.9 k high-pass (4 kHz) feeds an OPA810 at x100 (1.4 MHz: the transducer's
own resonance sets the band). Time of flight at 343 m/s: 5.8 us per mm there and back. Put the two transducers
in pins 1 and 5 (10.16 mm: their leads are 10 mm apart).

A12 audio

tutorial 1.05, 6.04 (+−5 V)

cells (2,3), (3,3)

EXPECT: a speaker on SPK follows AUDIO IN (2.5 W into 4 ohm) or the one-bit stream on Pmod IO2 (10 k / 10 n): MIC OUT is the electret x100
PAM8302A: 1 u couples AUDIO IN to IN+ (IN− grounded through 1 u), the one-bit DAC on Pmod IO2 arrives through 10 k
into 10 n (1.6 kHz) at the same node, so both play. BYP 1 u, SD high: on with the tile. The electret (pin 1 is the
+ side, 2.2 k from VS) through 1 u into an OPA810 at x100: a few hundred millivolts for speech.

Drawn by scripts/gen_sch.py from boards/amod_a.py: the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
gain blocks, an HF LNA, Johnson noise, a one-pin ADC, a 74HC4046 PLL, heat, a mixer, light, ultraso
Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/gallicchio/
Sheet: /
File: amod_a.kicad_sch

Title: AMOD-A: active building blocks on USB-C power

Size: A1 | Date: 2026-10-25 | **Rev: A**
KiCad E.D.A. 10.0.7 | Id: 1/1